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New experiments and validated master-equation modeling for OH production in propyl + O_2 reactions

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Abstract

The propyl + O_2 reactions are important prototype reactions for general alkyl + O_2 systems, possessing the most important reactive channels of larger species while remaining small enough for accurate quantum chemistry calculations and theoretical kinetics. Previous experiments [8] that characterized the production of OH in the propyl + O_2 reactions could be only qualitatively modeled by rigorous master-equation calculations, leaving a deficient validation of, for instance, the critical $RO_2 \leftrightarrow QOOH$ isomerization pathways. This article revisits the combined experimental and modeling study of DeSain et al., using an improved experimental method to obtain time-resolved OH concentrations for pulsed-photolytic Cl-initiated oxidation of propane. The results are modeled with master-equation-based rate coefficients incorporated into a kinetic mechanism. Quantitative agreement can be obtained, providing experimental validation for the OH-producing pathways that are crucial for autoignition. The importance of the rigorous treatment of the kinetics for these highly pressure and temperature dependent reactions is emphasized, as well as the significance of the formally direct pathways such as chemical activation.
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Keywords: Propyl oxidation; Master-equation method; Formally direct pathways; Hydroxyl radical; Autoignition

1. Introduction

The reaction of alkyl radical, R, with molecular oxygen is a central process in low-temperature combustion chemistry, as it is a main route of alkane autoignition. The thermal efficiency and levels of emissions of new advanced compression ignition engines [1,2] depend sensitively on precise timing of the ignition kinetics. Therefore, it has become imperative to develop accurate models for the oxidation of various fuels for a wide range of temperature and pressure. Because the oxidation mechanism of large hydrocarbons is theoretically and computationally cumbersome [3], great effort has been placed in understanding the "combustion archetype" ethyl + O_2 and propyl + O_2 systems, in the hope of extending the principles of these simpler systems to larger ones.

The addition of O_2 to an alkyl radical proceeds without a potential barrier to form the alkylperoxy (RO_2) radical [4,5],

$$R + O_2 \leftrightarrow RO_2 \quad \text{(R1)}$$

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This radical can then isomerize to hydroperoxyalkyl species (QOOH) via internal hydrogen abstractions [6],

$$RO_2 \leftrightarrow QOOH \quad (R2)$$

or directly eliminate HO_2 [7]:

$$RO_2 \rightarrow HO_2 + \text{alkene} \quad (R3)$$

The QOOH species is weakly bound and rapidly isomerizes back to RO_2 or dissociates to OH or HO_2 products [8],

$$QOOH \rightarrow HO_2 + \text{alkene} \quad (R4)$$

$$QOOH \rightarrow OH + \text{cyclic ether} \quad (R5)$$

Scission of the O–O bond in reaction (R5) is energetically more favorable than that of the C–O bond in reaction (R4), with the result that the major product channel of QOOH decomposition is OH and cyclic ether [4]. The QOOH has never been observed directly, but its reaction with O_2,

$$QOOH + O_2 \rightarrow O_2QOOH \rightarrow \ldots \rightarrow 2OH + OQ'O \quad (R6)$$

is responsible for low-temperature chain branching [9]. Monitoring of OH formation is not only a benchmark for kinetic modeling, but is also a surrogate for the production of QOOH. Furthermore, another important characteristic of OH is its effectiveness as a chain carrier. Hence understanding the formation of OH in $R + O_2$ reactions is central to modeling the chain-propagation and chain-branching in low-temperature hydrocarbon oxidation [9].

An important aspect of modeling these reactions rigorously is that product formation in $R + O_2$ systems can occur via several different pathways. The OH radical can be produced by sequentially proceeding through stabilization and thermally activated isomerization steps in all intermediate wells (i.e., $R + O_2 \leftrightarrow RO_2 \leftrightarrow QOOH \rightarrow OH$, termed as sequential pathway), or without intermediate stabilization (e.g., the chemically activated reaction $R + O_2 \rightarrow OH +$ other products, termed as formally direct pathway), or with a mixture of the above processes (e.g., $R + O_2 \rightarrow QOOH \rightarrow OH$). The master-equation (ME) methodology of Miller and Klippenstein [10,11] takes into account both sequential and formally direct processes, and assigns phenomenological rate coefficients to them in a mathematically rigorous manner.

The ethyl + O_2 system is by far the most studied theoretically and experimentally (see DeSain et al. [8] and references cited therein), because it is the smallest system in which the $RO_2 \leftrightarrow QOOH$ isomerization plays an important role. However, the major disadvantage of this system as a prototype reaction is that the transition state corresponding to reaction (R2) has a higher potential energy than do the reactants R and O_2 [4], a characteristic that is atypical of $R + O_2$ reactions of the larger alkanes [12]. A consequence of this fundamental difference is that the ethyl + O_2 system produces substantially less OH (via the QOOH intermediate) than do larger alkyl + O_2 counterparts [6], where the corresponding $RO_2 \leftrightarrow QOOH$ transition states include a more energetically favorable six-member ring. Hence, the reactions of propyl + O_2 or perhaps butyl + O_2 [12,13] are more realistic combustion "archetypes". The study of DeSain et al. [8,14] covers many of the details of the propyl + O_2 system at low pressure (~20 Torr) in the temperatures ranging from 298 to 700 K.

The present study is largely a reinvestigation of the work of DeSain et al. [8,14] on the propyl + O_2 system, in an effort to remedy the disparities that remained at its conclusion. Specifically, although the model reproduced behavior qualitatively similar to the experimental results and accurately predicted the time behavior of OH production, the model could not reproduce the temperature dependence of the OH yield, substantially under-predicting the observed OH concentrations at higher temperatures and over-predicting at lower temperatures. Other investigators [15] have also found it difficult to quantitatively model the experimental yields of DeSain et al. [8,14]. The principal difficulties in interpretation of the earlier experiments are the need to calibrate the LIF detection and the unwanted side reactions of the $CFCl_2$ co-product of the photolytic production of Cl atom from $CFCl_3$ photolysis. This study corrects both of those deficiencies, employing a cleaner source of chlorine radicals, oxalyl chloride $(COCl)_2$ [16] photolysis, to initiate the oxidation, and ultraviolet direct absorption to measure absolute OH concentrations. As in the study of DeSain et al. [8,14], time-dependent multiple-well master-equation calculations are used to derive parameterized rate coefficients for the R (n-C_3H_7 and i-C_3H_7) + O_2 systems, following the methods of Miller and Klippenstein [10,11]. These rate coefficients, along with literature rate coefficients for other reactions where needed, are incorporated into a rate-equation model of the Cl-initiated oxidation of propane. These computational results are then compared with the experimentally observed OH formation at 10 Torr, over temperatures from 400 to 700 K. Excellent agreement is obtained between the theory and new experimental data. Comparisons with previous results and literature are also made.

2. Experiment

In the previous study by DeSain et al. [8,14], Cl atoms were generated by photolysis of CCl_3F at 193 nm,

$$CCl_3F + h\nu\ (193\ \text{nm}) \rightarrow Cl + CFCl_2. \quad (R7)$$

Then propyl oxidation was initiated by the H abstraction from propane by Cl atom. The time-resolved OH signal in the reactions was probed by laser-induced fluorescence (LIF), which, although highly sensitive, requires calibration to yield absolute concentrations. The measured OH signal in the study of DeSain et al. was calibrated by measuring OH produced from the reference system, $CH_3OH/O_2/NO$ [8], which was also Cl-initiated through photolysis (R7) and involved three chemical conversion steps ($Cl \rightarrow CH_2OH \rightarrow HO_2 \rightarrow OH$). More importantly, the $CFCl_2$ co-product can react with the dominant HO_2 product of the $R + O_2$ reaction to give OH. This reaction was accounted for in the model of DeSain et al. [8,14], but its rate constant and branching fraction are uncertain, and it could produce a substantial fraction of the observed OH.

In order to avoid the above-mentioned difficulties with this radical precursor, in this study we generate Cl atoms through the photolysis of oxalyl chloride $(COCl)_2$ at 266 nm [16],

$$(COCl)_2 + h\nu\ (266\ \text{nm}) \rightarrow 2CO + 2Cl. \qquad \text{(R8)}$$

The $(COCl)_2$ is a much cleaner Cl source because the byproduct CO is not reactive. Although the $(COCl)_2$ photolysis produces some COCl product at 266 nm [16], COCl is very weakly bound [17] and will rapidly dissociate already at 400 K. In the present study, we use a differential direct absorption method at 308 nm to measure the time-resolved OH formation [18,19]. Direct absorption method avoids signal calibration because the absorption cross section of the OH line probed is well known [20,21]. We use the $Q_1(1)$ line of the $A^2\Sigma^+, v' = 0 \leftarrow X^2\Pi, v'' = 0$ absorption band of OH at 32474.58 cm^{-1} [20,21] to probe the OH concentration. The absolute number density of OH is calculated from the absorption signal using the method of Dorn et al. [18,21].

This 308 nm probing UV light is generated by frequency doubling the 616 nm output of the CW ring dye laser, whose output is actively locked to the high finesse cavity of the doubler to improve the doubling efficiency. The typical UV power in our experiments is about 20 mW. A Herriott-type multi-pass, slow-flow reactor cell [22,23] is used to improve the signal-to-noise ratio of the OH signal. About 10% of the probe beam is extracted out with a beam splitter before the cell and is used as a reference beam. The remaining 90% passes through the 1.5 m long Herriott-type cell which is inside the quartz reactor [19]. Two matched high-speed silicon detectors monitor these two beams. Their signals are subtracted and amplified in a low-noise differential amplifier. The DC amplitudes of the reference signal and the main signal are balanced, that is, the output of the differential amplifier is zero when there is no OH absorption in the cell, largely cancelling fluctuations in the probe laser power and improving sensitivity. The time-resolved differential absorption signal is averaged over about 100 photolysis laser pulses. For the present system, the minimum detectable OH concentration is about $1 \times 10^{10}\ cm^{-3}$.

The photolysis laser is the fourth harmonic of a pulsed Nd:YAG laser (266 nm), which passes directly through the center of the cell along its axis. The beam diameter is about 7 mm with typical photolysis energy of about 20 mJ per pulse. The probe UV laser makes 25 passes through the cell, overlapping with the photolysis beam at the center, where the pertinent reactions are probed and the temperature can be stabilized [23]. The typical absorption path length in this probing region is about 10 meters. With this direct absorption method, the time-dependent OH density, $[OH](t)$, can be easily measured.

The overall dissociation yields two chlorine atoms per photolyzed molecule (~2–5% of the precursor in the present experiments) on a time-scale <1 μs. This is essentially instantaneous for the kinetics investigated here; the Cl atom then reacts with the fuel molecules to give the alkyl radicals,

$$Cl + RH \rightarrow HCl + R, \qquad \text{(R9)}$$

which then initiates reaction (R1). A sufficient concentration of fuel (RH) is supplied so that the Cl atoms are converted into alkyl radicals quantitatively and rapidly (typically in less than 10 μs), minimizing secondary reactions of Cl atom. Concomitantly, the O_2 concentration is held considerably larger (at least by a factor of 30) than that of the fuel, such that the $R + O_2$ reaction is the dominant reaction involving the alkyl radical. Oxalyl chloride (99+%) was subjected to several freeze–pump–thaw cycles before use. Typical initial concentrations were $[(COCl)_2] \approx 5 \times 10^{14}\ cm^{-3}$, $[C_3H_8]$ ($\geqslant$99.999%) $\approx 5 \times 10^{15}\ cm^{-3}$, $[O_2]$ ($\geqslant$99.9999%) $\approx 2 \times 10^{16}\ cm^{-3}$, with He ($\geqslant$99.9999%) added to reach the total pressure of 10 Torr, maintained by a throttle valve with feedback loop. The individual gas flows are set by calibrated mass flow controllers and the total flow rate in the experiments is large enough to refresh the cell after each photolysis pulse at a 1 Hz repetition rate. A liquid-nitrogen-cooled trap is used in the reactor outlet to prevent back-streaming from the mechanical pump.

3. Theory and kinetic modeling

In order to validate rate coefficients and branching ratios from the observed OH concentration–time profiles, a chemical reaction mechanism must be constructed. The experimental data obtained in this work is modeled using ME calculations for the rate coefficients of the propyl + O_2 systems [8,12,14] embedded in a small chemical mechanism that includes initiation

reactions and secondary chemistry. The $C_3H_7 + O_2$ model uses 12 ME reactions for both the *n*- and *i*-propyl + O_2 components (with the ratio *n*-/*i*- close to 50%), combined with 51 auxiliary reactions (see Supplementary data). Sensitivity analysis (see below) shows that the auxiliary reactions do not significantly contribute to the observed OH concentration profiles under the present experimental conditions. However, in order to unambiguously identify OH production sources, the mechanism was not reduced. The auxiliary reactions considered are identical to those used by DeSain et al. [8,14], except of course that the reactions involving CCl_2F radicals have been eliminated. The master-equation rate coefficients at 10 Torr are provided as Supplementary data.

Figure 1 shows the most important features of the PES of the reactions studied, adopted from the paper of DeSain et al. [8]. The *i*-propyl + O_2 PES is very similar to that of ethyl + O_2 [4] with the crucial exception that all RO_2 isomerization barriers lie below the entrance asymptote. The isomerization of the *n*-propylperoxy radical can lead to two different QOOH species, with the radical center on either the central (CH_3CHCH_2OOH) or on the terminal ($CH_2CH_2CH_2OOH$) carbon. However, although the formation of the latter structure is facile, it plays only a minor role in the present kinetics, because the barrier leading to bimolecular products is considerably higher even than the reactants. Above ~450 K this QOOH cannot be stabilized at 10 Torr because of its rapid equilibrium with RO_2, and merges with RO_2 in a kinetic sense (although with a different reactivity, e.g., with O_2). Although this QOOH species is nearly irrelevant at the conditions of these experiments [8,11,14], under other conditions, such as higher pressure, it may have a significant role.

The stationary-point characteristics, including the optimized geometries and frequencies, are

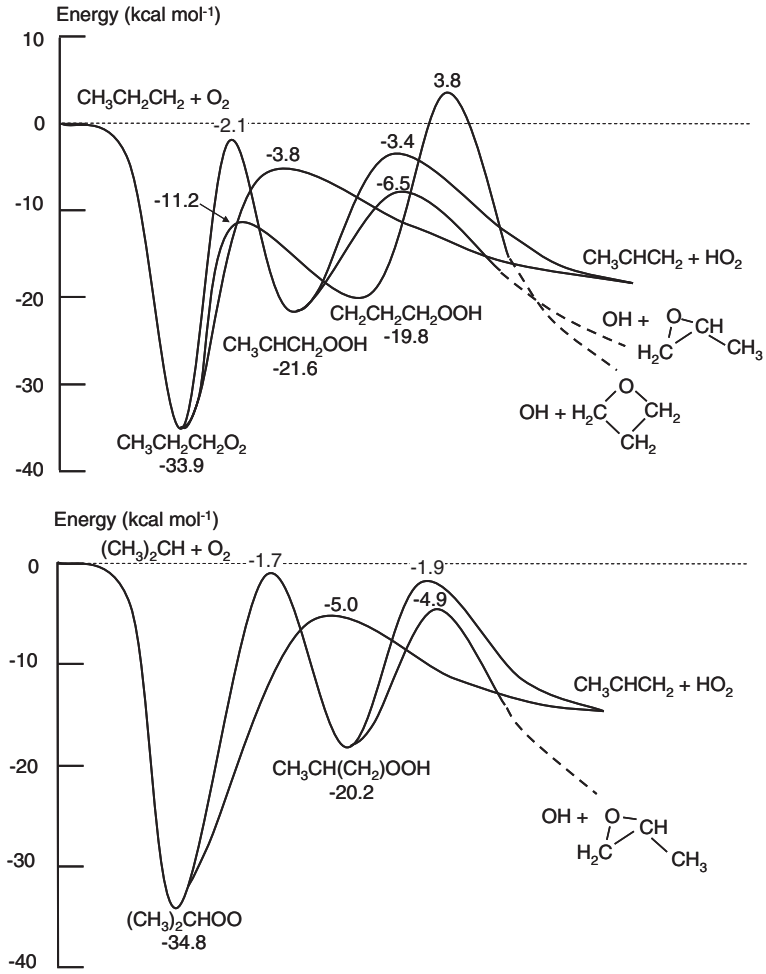

Fig. 1. Schematic potential energy surfaces of the most important channels in, *n*-propyl + O_2 (top) and *i*-propyl + O_2 (bottom) reactions. Numbers indicate the signed energies (including ZPE) of the barriers and well depths in kcal mol^{-1} used in this work. The barrier heights have been adjusted for best fitting.

adapted from the work of DeSain et al. [8]. The geometries and frequencies were obtained at the B3LYP/6-31G* level, while energies are from QCISD(T)/6-311++G(3df,2pd) using MP2 corrections [12]. The variable-reaction-coordinate transition-state-theory treatment of the barrierless entrance channels was also identical to that described in the paper of DeSain et al. [8] and in previous works [12]. The rate coefficients were calculated using the Variflex code, version 2.0 [24], and the Kintecus program [25] was used for kinetic modeling. The Supplementary data contains Arrhenius parameters for all of the reactions in the propyl + O_2 system for 10, 100, 760, and 7600 Torr for the 300–750 K range.

4. Results and discussion

Figure 2 shows the experimental and modeled OH concentrations normalized by the initial Cl atom concentration; in this way the change in the OH amplitude is shown independent of the slight variation in the initial Cl atom concentration $[Cl]_0$. The value of $[Cl]_0$ varies due to changes in the photolysis laser power as well as in the initial $(COCl)_2$ concentration. The estimated uncertainty in $[Cl]_0$ is 30%, based on the quality of the photolysis laser beam profile, uncertainty in the absorption cross section and concentration of $(COCl)_2$, and other minor factors. Within the uncertainties, the model behaves nearly linearly with respect to changes in $[Cl]_0$: more chlorine atoms produce proportionally more OH radicals, without significantly changing the time behavior of the OH concentration.

Figure 2 shows that [OH] increases as a function of temperature while the peak of the OH is shifted to longer times. Our modeling results are in remarkable agreement with the experimental ones for the propyl + O_2 reaction, substantially better than the earlier experiments in which CCl_3F molecule was used as a radical source [14]. In order to achieve the "best fit", the barrier between RO_2 and the corresponding hydroperoxypropyl QOOH for both radical isomers was lowered by 0.25 kcal mol^{-1} relative to the value used by DeSain et al. [14]. Because both radical isomers contribute significantly to OH production and because of the high degree of similarity in PES structure, changes were made equally to both PES's. This slight modification is still well within the uncertainty of the quantum chemical calculations.

Building on the very good agreement between model and experiment, the processes affecting the OH formation are assessed in order to determine to which features of the PES the model is the most sensitive, and which features are not tested under the current conditions. This is a very important aspect for predicting the reliability of the current validated stationary-point characteristics when applied at more practically important pressures (tens of bar). Fernandes et al. [26] demonstrated that validation of stationary points at low pressure can provide accurate predictions at higher pressures. For this reason, we calculated local sensitivity coefficients of the modeled OH concentration. Figure 3 presents the local, fully normalized sensitivities, $\partial \ln[OH]/\partial \ln k_i$ [27] in the propyl + O_2 system at 500 K. The analysis indicates that in the propyl + O_2 systems at 10 Torr and at temperatures in the 400–700 K range, the formation of the hydroxyl radical is most sensitive to the formally direct pathways, with the stepwise, well-to-well pathways playing a more minor role under these conditions. Of course the transition state characteristics for the

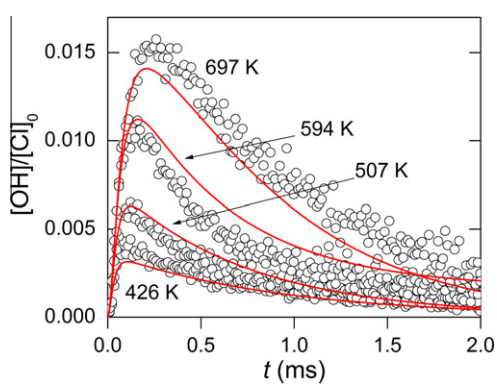

Fig. 2. Measured (circles) and modeled (lines) OH concentrations normalized by the initial chlorine atom concentration in the propyl + O_2 systems at various temperatures.

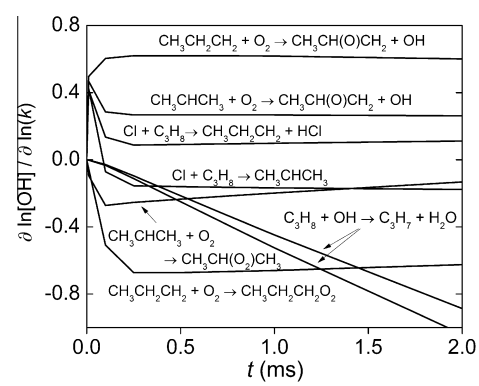

Fig. 3. Fully normalized sensitivities of the calculated OH concentration with respect to the rate coefficients for the propyl + O_2 system at 507 K. Only the most important reactions are shown. No reactions appear among the most sensitive reactions other than the "minimum set": formation of propyl radicals, their reaction with O_2, and the removal of OH by reaction.

$RO_2 \leftrightarrow QOOH$ isomerization are also manifested in the formally direct pathways.

To investigate the sensitivity of the present measurements to the stationary point energies, the OH concentration was modeled using slightly altered barrier heights. Figure 4 illustrates the response of the system to adjustments of the $RO_2 \leftrightarrow QOOH$ and $QOOH \leftrightarrow OH + \text{products}$ barriers by ± 1 kcal mol^{-1}. Because the present experiments do not distinguish the OH production from the individual isomers, both barrier heights of isomerization were adjusted by the same amount on the two PES's. As shown, the modeled production of OH in the Cl-initiated propane oxidation system is more sensitive to changes in the barrier height between the RO_2 and QOOH isomers than that of the barrier between the QOOH species and the bimolecular products.

The isomerization of alkylperoxy radicals is a critical step towards chain branching and hence the accurate treatment of the isomerization is important for hydrocarbon oxidation models. DeSain et al. [28] called into question the accepted values for the six-membered ring isomerization states based on modeling of their experiments on neopentyl radical oxidation. Their model used a relatively rapid isomerization, which was drawn by analogy with the master-equation calculations of n-propyl + O_2. Although the present experiments are insensitive to the six-membered ring transition state, they provide validation for the five-membered ring transition states linking propylperoxy radicals with the corresponding 2-hydroperoxypropyl radicals. Figure 5 shows the values for the forward rate coefficient for the isomerizations of $RO_2 \leftrightarrow QOOH$ for both i-propyl

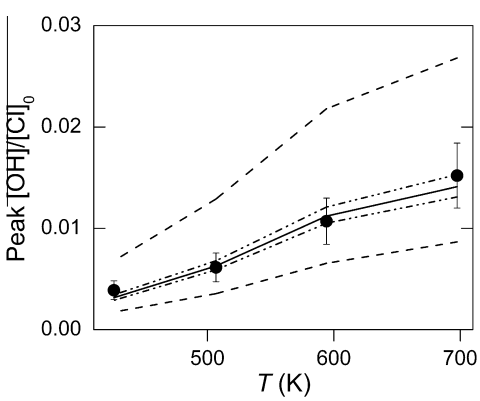

Fig. 4. Normalized experimental peak OH concentration as a function of temperature for the propyl + O_2 systems (●). The solid line corresponds to the "best fit" modeling results. The dashed lines show the product formation when the $RO_2 \leftrightarrow QOOH$ barrier is adjusted by ±1 kcal/mol and the dash-dot-dash lines show the response to adjusting that of $QOOH \leftrightarrow OH + c\text{-}C_3H_6O$ by ±1 kcal/mol. Reducing the barriers results in more OH, while increasing them results in less.

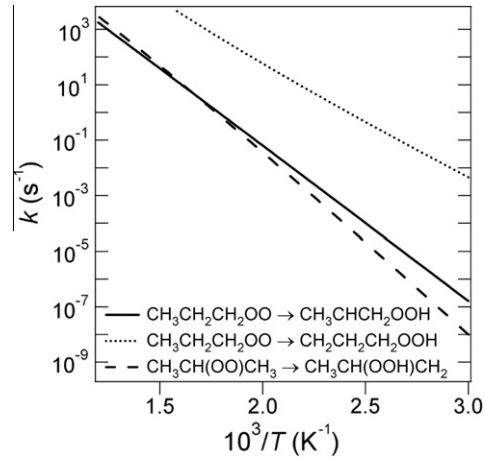

Fig. 5. Calculated rate coefficients for isomerization of propylperoxy radicals at 7600 Torr total pressure.

and n-propyl isomers, calculated at 7600 Torr total pressure (helium bath gas). The calculated rate coefficients for isomerization over the five-membered ring transition states are in reasonable agreement with values employed in comprehensive models [29], with a slightly smaller activation energy.

The six-membered ring isomerization forms the very weakly bound $CH_2CH_2CH_2OOH$ radical, which reaches its stabilization limit, that is, the temperature above which stabilization in the well is impossible at any pressure, below 500 K. Strictly speaking, the isomerization rate coefficient shown in Fig. 5 is valid only below the stabilization limit, but it is clear that the rate coefficient has a markedly higher A-factor and lower activation energy than the five-membered ring transition states. Above the stabilization limit the $CH_2CH_2CH_2OOH$ species is incorporated into the $CH_3CH_2CH_2OO$ well, and the master-equation methodology calculates unimolecular reactions from the $CH_2CH_2CH_2OOH$ configuration, for example, formation of oxirane + OH products, effectively as formally direct reactions from $CH_3CH_2CH_2OO$. The present model leaves unchanged the stationary points responsible for the HO_2-forming channels in the propyl + O_2 reaction, so the agreement with experiment for those channels [30,31] is retained. As a result, the present model represents a widely-validated set of stationary points that can be used to predict rate coefficients at many different conditions.

5. Conclusions

This paper re-examined the ab initio master-equation model describing the propyl + O_2 reactions by comparison to new sensitive long-path

absorption experiments that measured the time-dependence of OH radical formation under controlled conditions. These new low-pressure experiments are principally sensitive to the long-range part of the potential governing rate constants for $R + O_2$ addition, to the $RO_2 \leftrightarrow QOOH$ isomerization and $QOOH \leftrightarrow OH +$ cyclic ether dissociation barriers, and indirectly to the energy transfer processes. The remarkable agreement between experiment and model serves to further validate the theoretical understanding of this prototypical system. Further experiments, perhaps at high pressure or probing other species, may be useful to test aspects of the potential energy surface, such as the $CH_2CH_2CH_2OOH$ formation, to which the present measurements are insensitive.

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Appendix A. Supplementary data

Master-equation calculated Arrhenius parameters and theoretical equilibrium constants for reactions in the propyl $+ O_2$ systems. Note that in the *n*-propyl $+ O_2$ case $CH_2CH_2CH_2OOH$ is a chemically distinct species only at the two higher pressures. Therefore, at 10 and 100 Torr 12, whereas at 760 and 7600 Torr 20 rate coefficients are presented. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.039.

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